

Chapter 10: Gibbs Free energy Composition and Phase Diagrams of Binary Systems

Part II

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Behavior in the Liquid and Solid States

10.4 Liquid And Solid Standard States

- ✓ The **choice** of the **standard state** of a component of a condensed system is often that of the **pure component** in its **stable state** at the particular temperature and pressure of interest.
- ✓ At **1 atm** pressure (the pressure normally considered), the stable state is determined by whether or not the temperature of interest is above or below the normal **melting temperature** of the component.
- ✓ It is often **assumed** that the temperature of interest is either above or below the **melting temperatures** of **both** components.

10.4 Liquid And Solid Standard States

- ✓ An activity versus composition curve could be drawn for liquid immiscibility, in which case the standard states are the two pure liquids, or it could be drawn for solid immiscibility, in which case the standard states are the two pure solids.
- ✓ Since the standard state of a component is simply a reference state with which the component in any other state is compared, it follows that any state can be chosen as the standard state, and the choice is normally made purely on the basis of convenience.
- ✓ Consider the binary system $A-B$ at the temperature T , which is below the melting temperature of B , $T_{m(B)}$, and above the melting temperature of A , $T_{m(A)}$. Consider, further, that this system forms Raoultian ideal liquid solutions and Raoultian ideal solid solutions.

10.4 Liquid And Solid Standard States

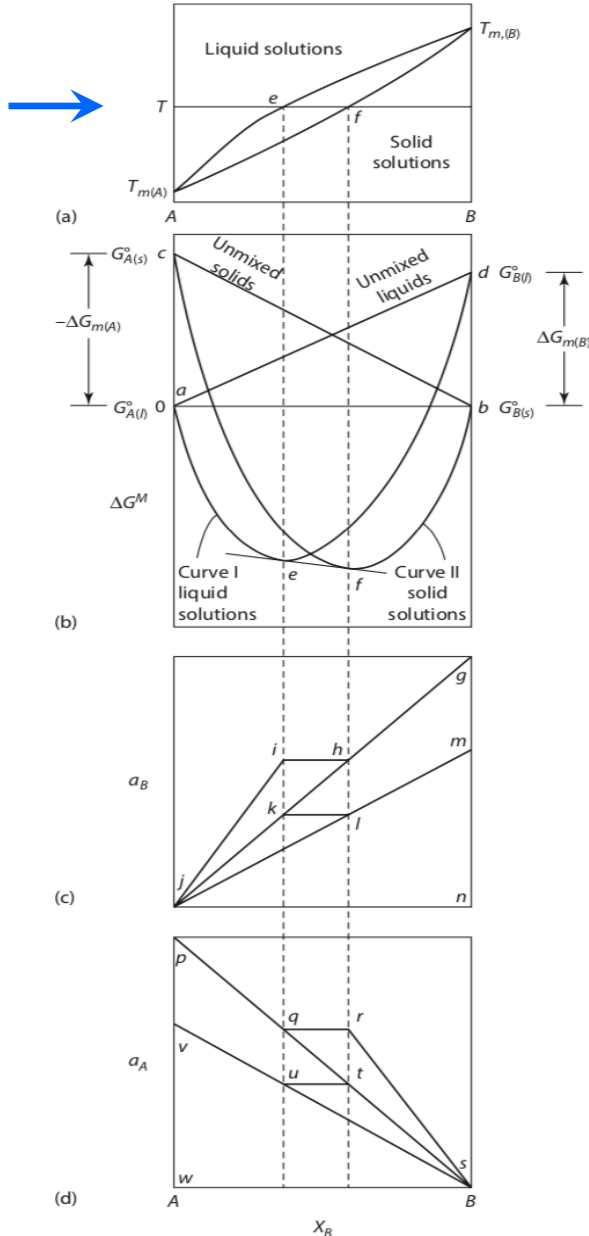


Figure 10.8

(a) the phase diagram for the system A–B.

(b) the Gibbs free energies of mixing in the system A–B at the temperature T .

(c) the activities of B at the temperature T and comparison of the solid and liquid standard states.

(d) the activities of A at the temperature T and comparison of the solid and liquid standard states.

10.4 Liquid And Solid Standard States

- ✓ The phase diagram for the system is shown in **Figure 10.8a**. **Figure 10.8 b** shows the two Gibbs free energy of mixing curves at the temperature of interest. **Curve I** is drawn for liquid solutions and **curve II** is drawn for solid solutions.
- ✓ At the temperature T , the stable states of pure A and B are located at $\Delta G_M = 0$, with pure liquid A located at $X_A = 1$ (the point **a**) and pure solid B located at $X_B = 1$ (the point **b**). The **point c** represents the molar Gibbs free energy of solid A relative to that of liquid A at the temperature T , and $T > T_{m(A)}$. Thus $G_{A(s)}^\circ - G_{A(l)}^\circ = -\Delta G_{m(A)}$ is a **positive** quantity which is equal to the negative of the molar Gibbs free energy of melting of A at the temperature T . That is,

$$G_{A(s)}^\circ - G_{A(l)}^\circ = -\Delta G_{m(A)}^\circ = -(\Delta H_{m(A)}^\circ - T\Delta S_{m(A)}^\circ)$$

10.4 Liquid And Solid Standard States

- ✓ and if $c_{PA(s)} = c_{PA(l)}$ (i.e., if $\Delta H_{m(A)}$ and $\Delta S_{m(A)}$ are independent of temperature), then

$$\Delta G_{m(A)}^{\circ} = \Delta H_{m(A)}^{\circ} \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right)$$

- ✓ As the temperature decreases and approaches the melting point of A, $\Delta G_{m(A)}^{\circ}(T)$ approaches **zero**.
- ✓ Similarly, the point **d** represents the molar Gibbs free energy of **liquid B** relative to that of **solid B** at the temperature T , and, since $T < T_{m(B)}$, then $G_{B(l)}^{\circ} - G_{B(s)}^{\circ} = \Delta G_{m(B)}^{\circ}$ is a **positive** quantity.

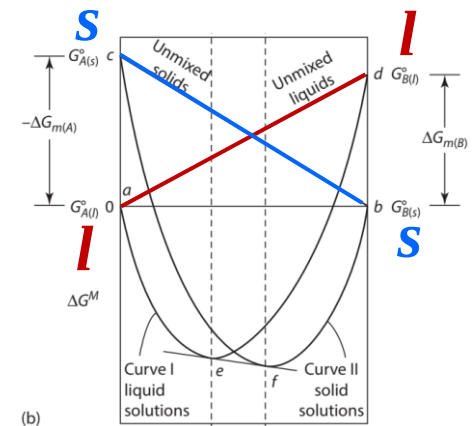
10.4 Liquid And Solid Standard States

- ✓ The line ad in **Figure 10.8b** joining a and d represents the Gibbs free energy of unmixed liquid A and liquid B relative to that of the standard state of unmixed liquid A and solid B , and
- ✓ the line cb joining c and b represents the Gibbs free energy of unmixed solid A and solid B relative to that of the standard state.
- ✓ The line cb can be represented by the equation

$$\Delta G = -X_A \Delta G_{m(A)}^{\circ}$$

- ✓ and the equation for the line ad is

$$\Delta G = X_B \Delta G_{m(B)}^{\circ}$$



10.4 Liquid And Solid Standard States

✓ At any composition, the formation of a homogeneous liquid solution from **liquid A** and pure **solid B** can be considered as being a two-step process involving

1. The **melting** of X_B moles of B , which involves the change in Gibbs free energy

$$\Delta G = X_B \Delta G_{m(B)}^{\circ}$$

2. The **mixing** of X_B moles of liquid B and X_A moles of liquid A to form an ideal liquid solution, which involves the change in Gibbs free energy

$$\Delta G = \Delta G^{M,id} = RT(X_A \ln X_A + X_B \ln X_B)$$

10.4 Liquid And Solid Standard States

- ✓ Thus, the molar Gibbs free energy of mixing (which can also be called the molar Gibbs free energy of formation) of an **ideal liquid** solution, $\Delta G_{(l)}^M$, from liquid A and solid B is given by

$$\Delta G_{(l)}^M = RT(X_A \ln X_A + X_B \ln X_B) + X_B \Delta G_{m(B)}^\circ$$

which is the equation of **curve I** in **Figure 10.8b**.

- ✓ Similarly, at any composition, the formation of an **ideal solid** solution from liquid A and solid B involves a change in Gibbs free energy of

$$\Delta G_{(s)}^M = RT(X_A \ln X_A + X_B \ln X_B) - X_A \Delta G_{m(A)}^\circ$$

which is the equation of **curve II** in **Figure 10.8b**.

10.4 Liquid And Solid Standard States

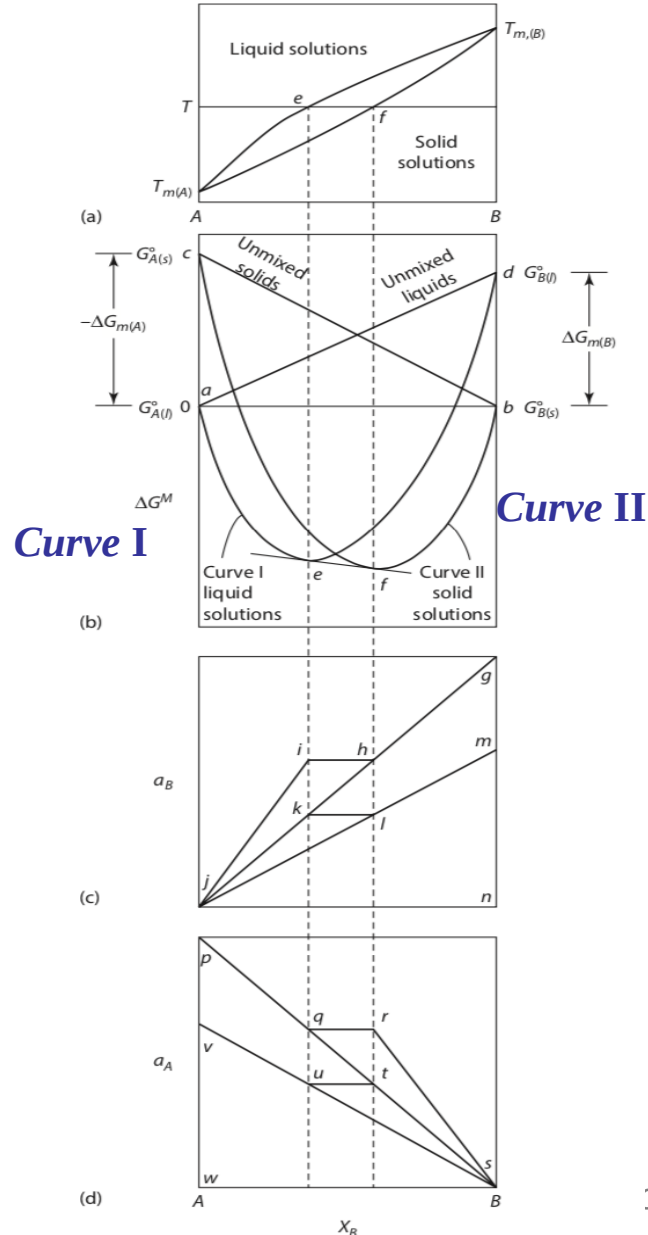


Figure 10.8

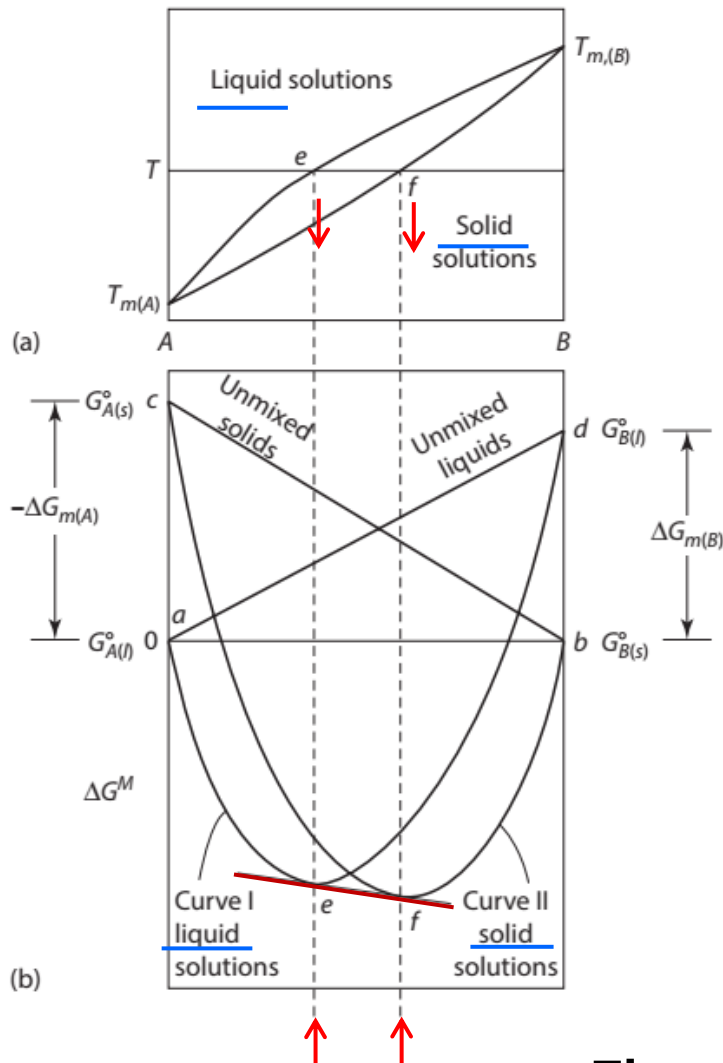
(a) the phase diagram for the system A-B.

(b) the Gibbs free energies of mixing in the system A-B at the temperature T .

(c) the activities of B at the temperature T and comparison of the solid and liquid standard states.

(d) the activities of A at the temperature T and comparison of the solid and liquid standard states.

10.4 Liquid And Solid Standard States



✓ At the composition e , the tangent to the free energy of mixing curve for the liquid solution is also the tangent to the free energy of mixing curve of the solid solution at the composition f .

✓ Thus, at the temperature T , the liquid of composition e is in equilibrium with the solid of composition f ; that is, e is the liquidus composition and f is the solidus composition, as seen in Figure 10.8a.

Figure 10.8

10.4 Liquid And Solid Standard States

- ✓ As the temperature is lowered, the magnitude of ca decreases until, at the melting point of A , the length of ca equals zero. With decreasing temperature, the magnitude of db increases.
- ✓ The consequent movement of the positions of curves I and II relative to one another is such that the positions e and f of the double tangent to the curves shift to the left.

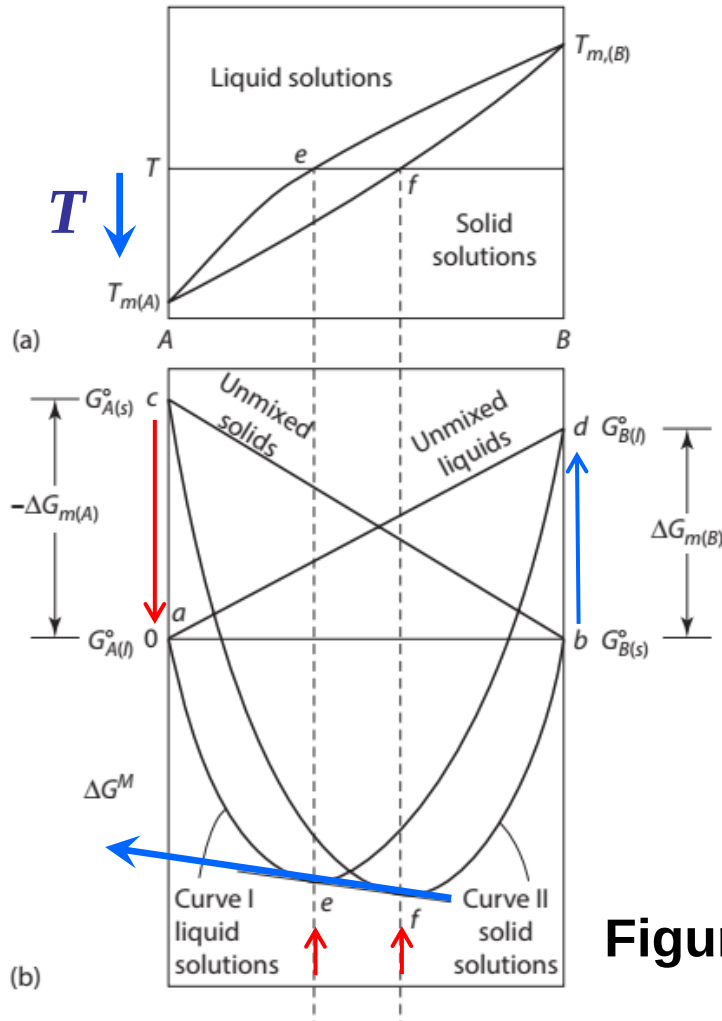


Figure 10.8

10.4 Liquid And Solid Standard States

- ✓ Correspondingly, if the temperature is increased, the relative movement of the Gibbs free energy curves is such that e and f shift to the right. The loci of e and f with change in temperature trace out the liquidus and solidus curves, respectively.
- ✓ The equations for the solidus and liquidus curves of this system are derived in Appendix 10A of this chapter. There, the equations are found to be

$$X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G_{m(B)}^{\circ}}{RT}\right)}{\exp\left(-\frac{\Delta G_{m(A)}^{\circ}}{RT}\right) - \exp\left(-\frac{\Delta G_{m(B)}^{\circ}}{RT}\right)}$$

10.4 Liquid And Solid Standard States

and

$$X_{A(l)} = \frac{\left[1 - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)\right] \exp\left(-\frac{\Delta G_{m(A)}^\circ}{RT}\right)}{\exp\left(-\frac{\Delta G_{m(A)}^\circ}{RT}\right) - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)}$$

✓ If $c_{p,i(s)} = c_{p,i(l)}$, we obtain from **Equation 10.1**, for $i = A$ and B ,

$$\Delta G_{m(i)}^\circ = \Delta H_{m(i)}^\circ \left[\frac{T_{m(i)} - T}{T_{m(i)}} \right]$$

✓ Thus, the phase diagram for a system which forms ideal solid and liquid solutions is determined **only by the melting temperatures** and the **molar heats** of melting of the components.

10.4 Liquid And Solid Standard States

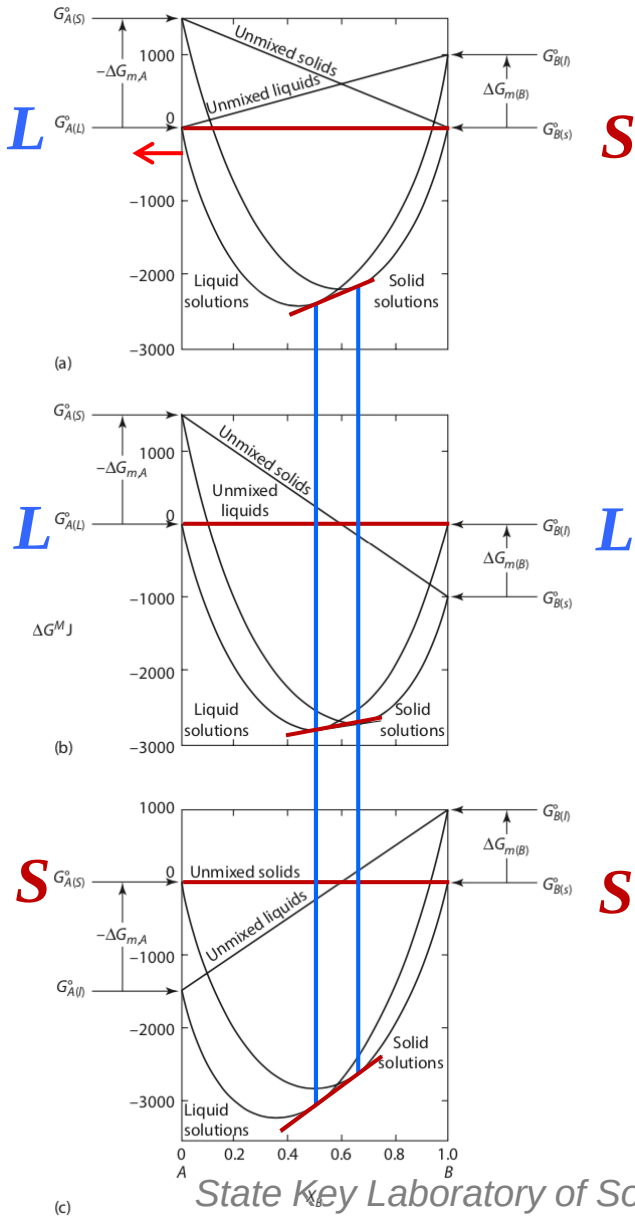


Figure 10.9

the Gibbs free energy of mixing curves for a binary system $A-B$ which forms **ideal solid** solutions and **ideal liquid** solutions, at a temperature which is higher than $T_{m(A)}$ and lower than $T_{m(B)}$.

(a) liquid A and solid B chosen as standard states located at $\Delta G^M = 0$.

(b) liquid *A* and liquid *B* chosen as standard states located at $\Delta G^M = 0$.

(c) solid *A* and solid *B* chosen as standard states located at $\Delta G^M = 0$.

the **positions** of the points of double tangency are **not** influenced by the choice of standard state.

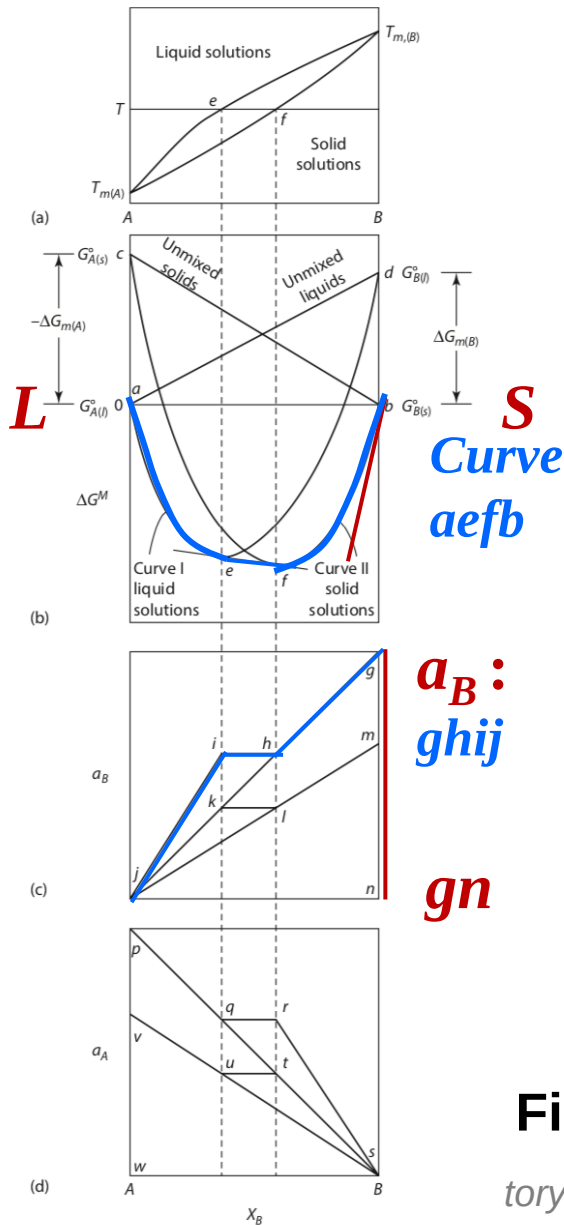
10.4 Liquid And Solid Standard States

- ✓ **Figure 10.9** shows the Gibbs free energy of mixing curves for a binary system A–B which forms ideal solid solutions and ideal liquid solutions, drawn at a temperature of 500 K, which is lower than $T_{m,(B)}$ and higher than $T_{m,(A)}$.
- ✓ At 500 K, $\Delta G^o_{m,A} = -1500 \text{ J}$ and $\Delta G^o_{m,B} = 1000 \text{ J}$.
- ✓ **Figure 10.9a** shows the curves when liquid A and solid B are chosen as the standard states, located at $\Delta G^M = 0$.
- ✓ **Figure 10.9b** shows the curves when liquid A and liquid B are chosen as the standard states, and
- ✓ **Figure 10.9c** shows the curves when solid A and solid B are chosen as the standard states.

10.4 Liquid And Solid Standard States

- ✓ Comparison among the three shows that, because of the logarithmic nature of the Gibbs free energy curves, the positions of the points of common tangency are not influenced by the choice of standard state; they are determined only by the temperature T and by the magnitude of the difference between $G^o_{(l)}$ and $G^o_{(s)}$ for both components at the temperature T .

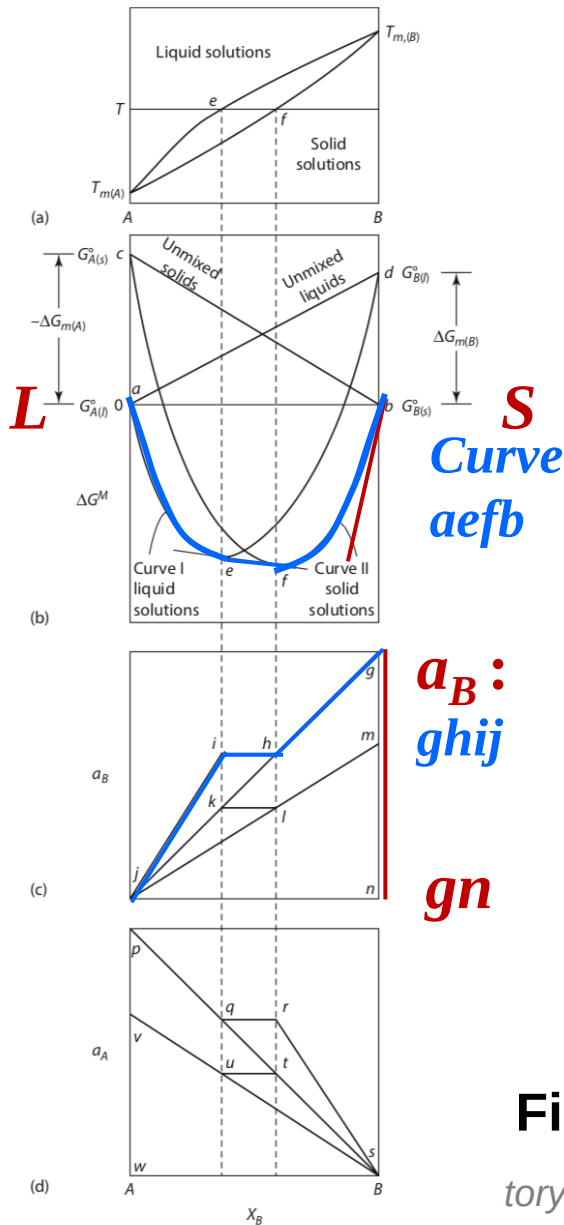
10.4 Liquid and Solid Standard States



- ✓ The activity–composition relationships for component B are shown in **Figure 10.8c**. There are two possible standard states (**Figure 10.8b**).
- ✓ The point b for solid B and the point d for liquid B could be chosen as standard states.

Figure 10.8

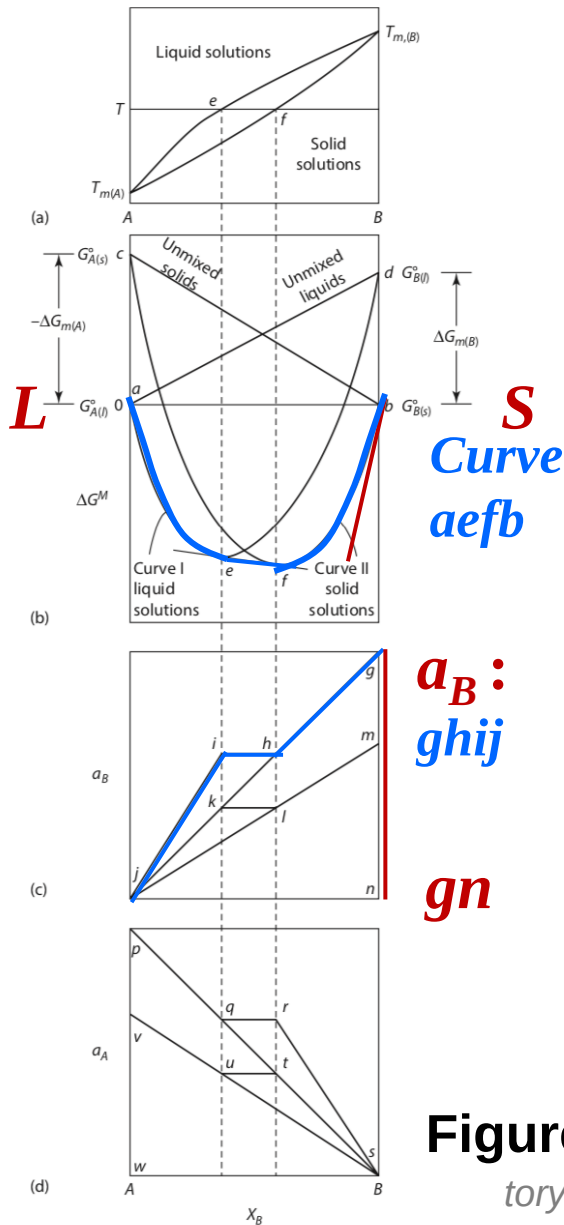
10.4 Liquid and Solid Standard States



- ✓ The lengths of the tangential intercepts with the $X_B = 1$ axis can be measured from b , in which case the activities of B are obtained with respect to solid B as the standard state,
- ✓ or
- ✓ the lengths can be measured from d , which gives the activities with respect to liquid B as the standard state.

Figure 10.8

10.4 Liquid and Solid Standard States



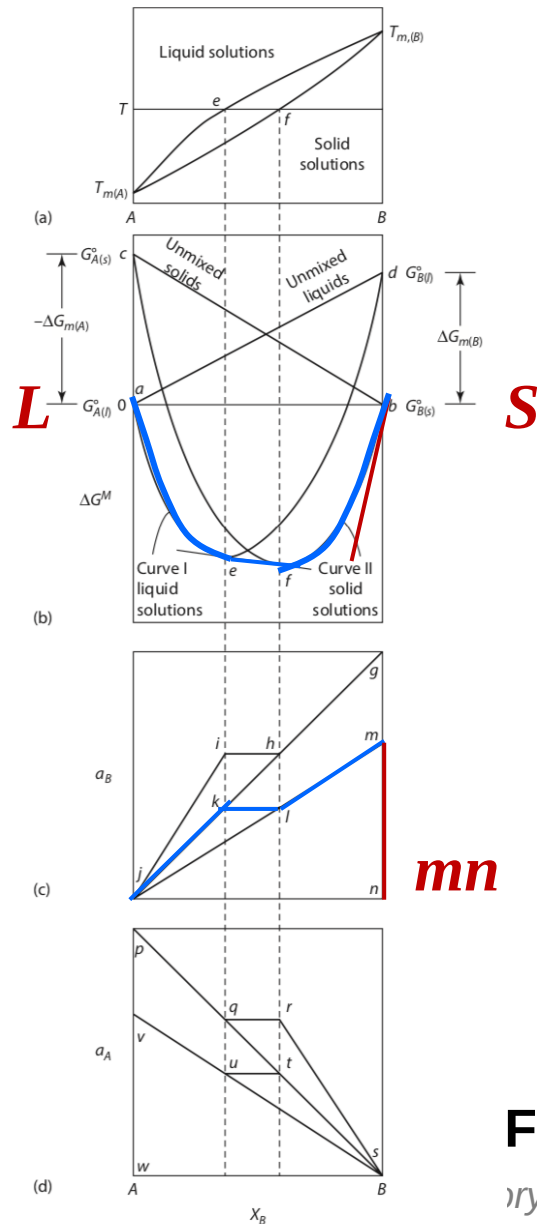
✓ If pure solid B is chosen as the standard state and is located at the point g in Figure 10.8c, then the length gn is, by definition, unity, and this defines the solid standard-state activity scale.

✓ The line $ghij$ then represents a_B in the solutions, with respect to solid B having unit activity at g .

✓ The line is obtained from the variation of the tangential intercepts from the curve $aefb$ to the $X_B = 1$ axis, measured from the point b .

Figure 10.8

10.4 Liquid and Solid Standard States



- ✓ On this activity scale, Raoult's law is given by ig , and the points i and h represent, respectively, the activity of B in the coexisting liquid solution e and solid solution f .
- ✓ The point m represents the activity of pure liquid B measured on the solid standard-state activity scale of B . This activity is less than unity, being given by the ratio mn/gn .

Figure 10.8

10.4 Liquid And Solid Standard States

- ✓ For B in any state along the *aefb* Gibbs free energy of mixing curve, in which state the partial molar Gibbs free energy of B is $\bar{G}_B$, the following relations hold.

$$\bar{G}_B = \underline{G_{B(l)}^\circ} + RT \ln (a_B \text{ with respect to } \underline{\text{liquid } B})$$

- ✓ and

$$\bar{G}_B = \underline{G_{B(s)}^\circ} + RT \ln (a_B \text{ with respect to } \underline{\text{liquid } B})$$

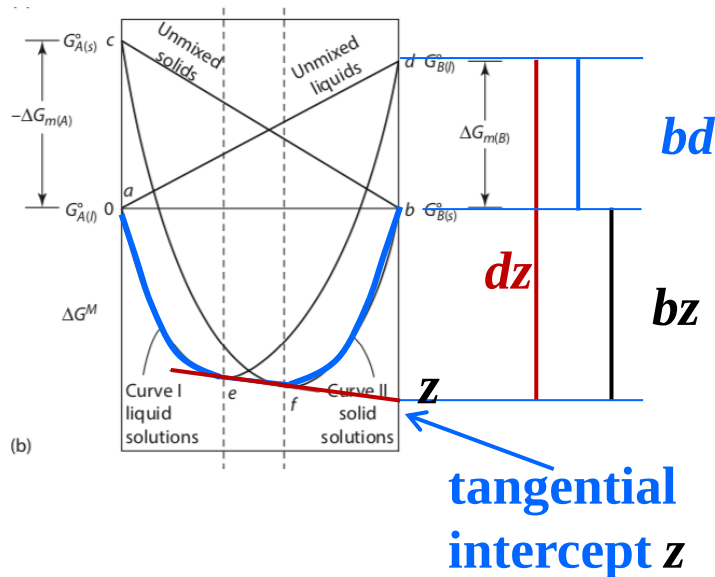
- ✓ Thus,

$$G_{B(l)}^\circ - G_{B(s)}^\circ = \Delta G_{m(B)}^\circ = RT \ln \left(\frac{a_B \text{ with respect to solid } B}{a_B \text{ with respect to liquid } B} \right)$$

10.4 Liquid And Solid Standard States

- ✓ Since $T < T_{m(B)}$, $\Delta G^0_{m(B)}$ is a positive quantity, the activity of B in any solution with respect to solid B as the standard state is less than the activity of B with respect to liquid B as the standard state, where both activities are measured on the same (solid or liquid) activity scale.
- ✓ For pure B , $a_{B(s)} > a_{B(l)}$ (i.e., $gn > mn$ in **Figure 10.3c**), and, if $gn = 1$, then $mn = \exp(-\Delta G^0_{m(B)}/RT)$.

10.4 Liquid and Solid Standard States

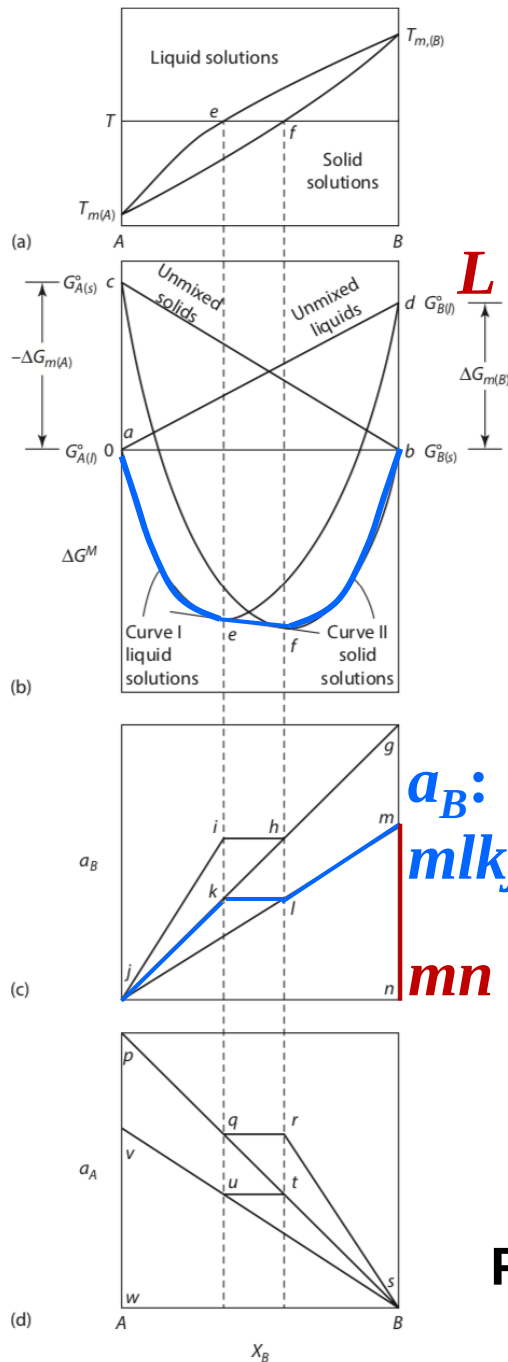


- **Equation 10.6** simply states that:
- the length of the tangential intercept from any point (i.e., ***f***) on the curve ***aefb***, measured from ***b*** (***bz***)
- added to the length ***bd***
- **equals**
- the length of the tangential intercept from the same point on the curve measured from ***d*** (***dz***)

$$dz = bd + bz$$

$$G_{B(l)}^{\circ} - G_{B(s)}^{\circ} = \underbrace{\Delta G_{m(B)}^{\circ}}_{bd} = RT \ln \left(\frac{\alpha_B \text{ with respect to solid B}}{\underbrace{\alpha_B \text{ with respect to liquid B}}_{dz}} \right)$$

uid and Solid Standard States



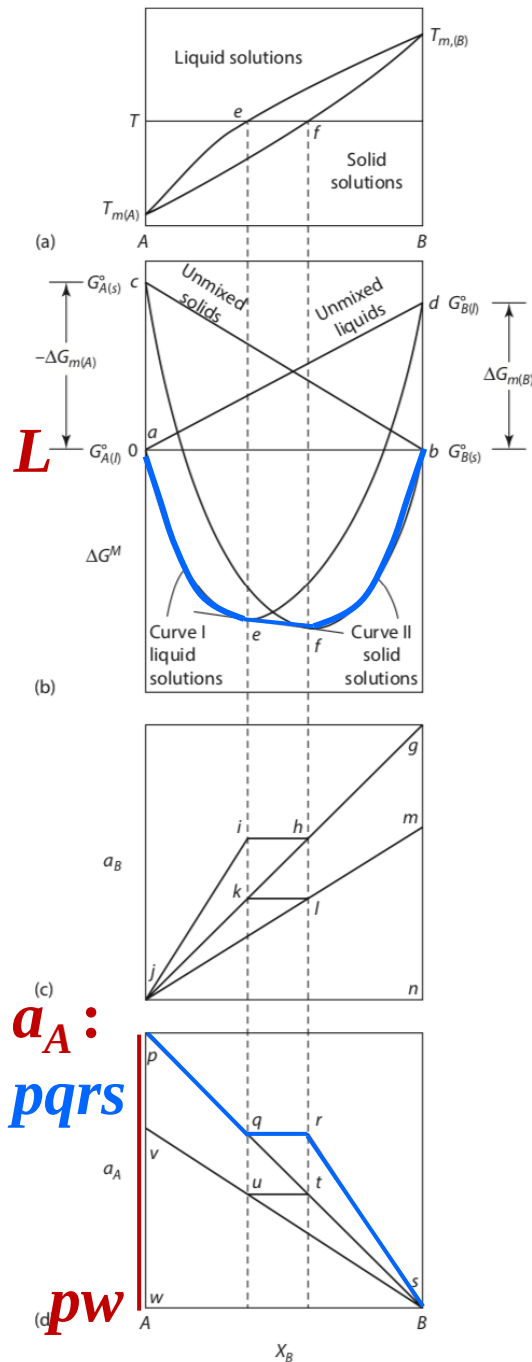
- ✓ **Figure 10.8c** shows $a_B \sim X_B$ for component B
- ✓ If pure liquid B is chosen as the standard state and is located at the point m , then the length mn is, by definition, unity, and this defines the liquid standard-state activity scale.
- ✓ Raoult's law on this scale is given by the line jm , and the **activities of B** in solution, with respect to pure liquid B having unit activity, are represented by the line $mlkj$.

Figure 10.8

10.4 Liquid And Solid Standard States

- ✓ The activity of solid B , located at g , is greater than unity on the liquid standard-state activity scale, being equal to $\exp(\Delta G^{\circ}m(B) / RT)$.
- ✓ When measured on one or the other of the two activity scales, the lines $jihg$ and $jklm$ vary in the constant ratio $\exp(\Delta G^{\circ}m(B) / RT)$,
- ✓ but
- ✓ $jihg$ measured on the solid standard-state activity scale is identical with $jklm$ measured on the liquid standard-state activity scale.

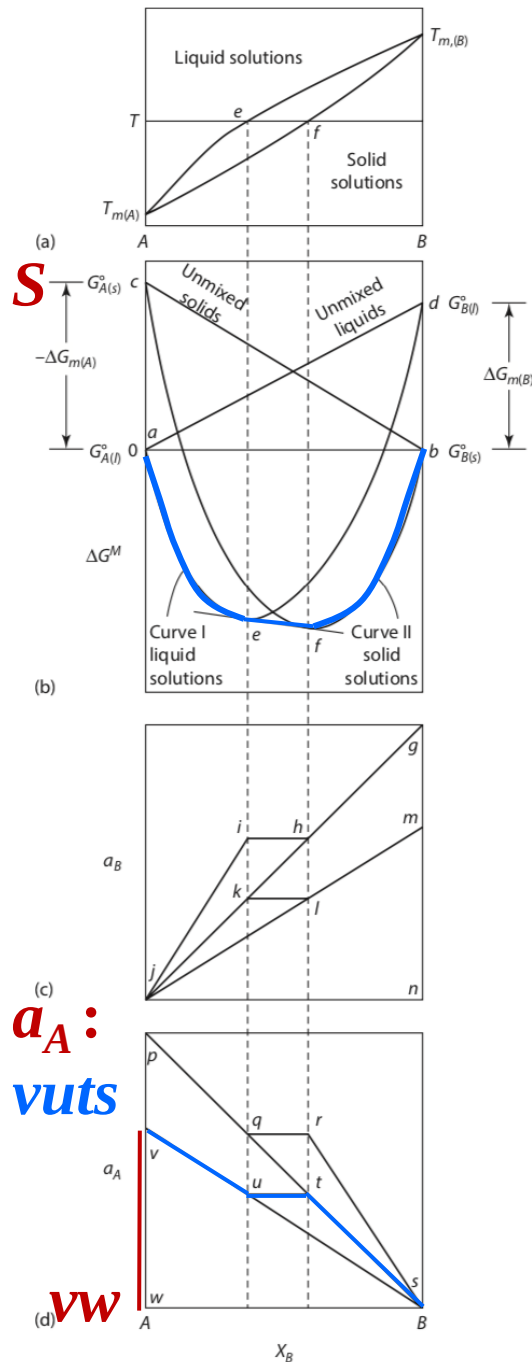
liquid and Solid Standard States



- ✓ **Figure 10.8d** shows $a_A \sim X_B$ for component **A**
- ✓ Since $T > T_{m(A)}$, $\Delta G^0_{m(A)}$ is a negative quantity, and hence,

$$\alpha_A(\text{with respect to liquid A}) > \alpha_A(\text{with respect to solid A})$$
- ✓ when measured on the same activity scale.
- ✓ If **pure liquid A** is chosen as the standard state and is located at the point **p**, then the length of **pw** is, by definition, unity, and the line **pqrs** represents the activity of **A** in the solution with respect to the liquid standard state.
- ✓ On the liquid standard-state activity scale, the activity of **pure solid A**, located at the point **v**, has the value $\exp(\Delta G^0_{m(A)} / RT)$.

uid and Solid Standard States



- ✓ If **pure solid A** is chosen as the **standard state**, then the length of **vw** is, by definition, unity, and
- ✓ Raoult's law is given by versus: **ps** .
- ✓ The line **$vuts$** represents the **a_A** in the solutions with respect to pure solid A.
- ✓ On the solid standard-state activity scale, liquid A, located at the **point p** , has the value $\exp(-\Delta G_{m(A)}^0 / RT)$. Again, the two lines, measured on one or the other of the two activity scales, vary in the **constant ratio** $\exp(\Delta G_{m(A)}^0 / RT)$, and when measured on their respective scales, they are **identical**.

10.4 Liquid And Solid Standard States

- ✓ If the temperature of the system is decreased to a value less than T indicated in **Figure 10.8a**, then the **length of ac** , being equal to $|\Delta G^0_{m(A)}|$ at the temperature of interest, **decreases**;
- ✓ correspondingly, the magnitude of $|\Delta G^0_{m(B)}|$, and hence the **length of bd** , **increases**.
- ✓ The consequent change in the positions of the Gibbs free energy of mixing **curves I and II** in **Figure 10.8b** causes the double tangent points **e** and **f** to **shift to the left toward A**.

10.4 Liquid And Solid Standard States

- ✓ The effect on the activities is as follows. In the case of both components,

$$\frac{a_i \text{ with respect to solid } i}{a_i \text{ with respect to liquid } i} = \exp\left(\frac{\Delta G_{m(i)}^\circ}{RT}\right)$$

which, from **Equation 10.1**,


$$\Delta G_{m(A)}^\circ = \Delta H_{m(A)}^\circ \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right)$$
$$= \exp \left[\Delta H_{m(i)}^\circ \left(\frac{T_{m(i)} - T}{RT T_{m(i)}} \right) \right]$$

10.4 Liquid and Solid Standard States

- ✓ With respect to component **B**,
 - if the temperature, which is less than $T_{m(B)}$, is decreased,
 - the ratio $a_{B(\text{solid})}/a_{B(\text{liquid})}$, which is greater than unity, increases.
- ✓ Thus, in **Figure 10.8c**, the ratio **gn/mn** increases.
- ✓ With respect to the component **A**,
 - if the temperature, which is higher than $T_{m(A)}$, is decreased,
 - then the ratio $a_{A(\text{solid})}/a_{A(\text{liquid})}$, which is less than unity, increases.
- ✓ Thus, the ratio **vw/pw** in **Figure 10.8d** increases.

- ✓ At the temperature $T_{m(A)}$, solid and liquid A **coexist** in equilibrium, $\Delta G^0_{m(A)} = 0$, and the points **p** and **v** coincide.
- ✓ Similarly, at the temperature $T_{m(B)}$, the points **m** and **g** coincide.

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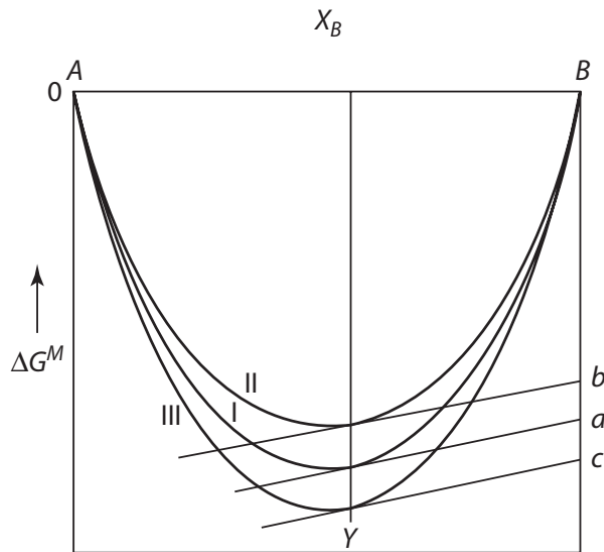
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- ✓ 10.2 Gibbs Free Energy and Thermodynamic Activity
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- ✓ 10.4 Liquid and Solid Standard States
- ✓ 10.5 The Gibbs Free Energy of Formation of Regular Solutions
- ✓ 10.6 Criteria For Phase Stability in Regular Solutions
- ✓ 10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity
- ✓ 10.8 The Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ Since curves II and III in Figure 10.1 are drawn for regular solutions, then the deviation of ΔG^M from $\Delta G^{M,id}$ is due only to the nonzero molar **enthalpy** of mixing and the difference between the two curves:

$$\Delta G^M - \Delta G^{M,id} = G^{XS} = \Delta H^M = \alpha X_A X_B$$

- ✓ For **curve II**, $|\Delta G^M| < |\Delta G^{M,id}|$, and thus, ΔH^M and α are positive quantities.

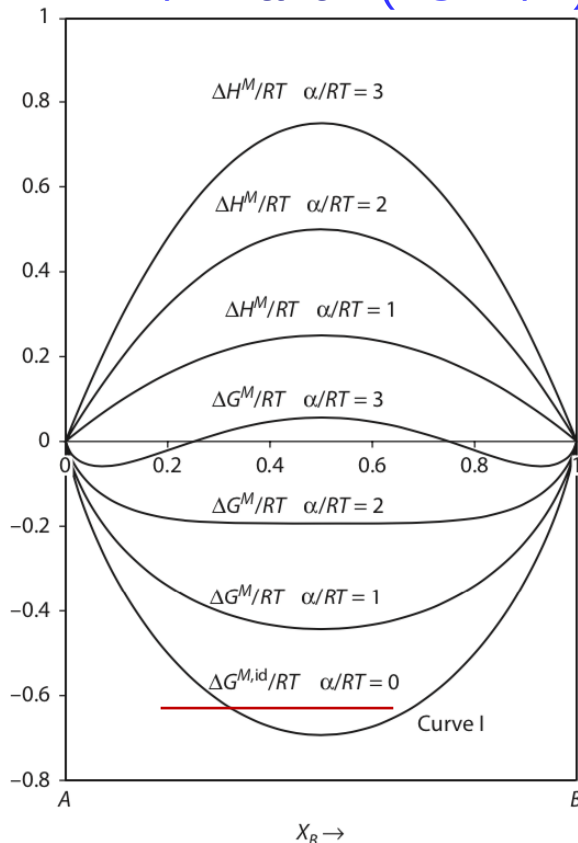


- ✓ It is of interest to consider the **effect of increasingly positive values of α on the shape of the Gibbs free energy of mixing curve for a regular solution.**

Figure 10.1 the molar Gibbs free energies of mixing in binary systems exhibiting **ideal behavior** (I), **positive deviation** from ideal behavior (II), and negative deviation from ideal behavior (III).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ In Figure 10.10, **curve I** is drawn as $-\Delta S^{M,id}/R = X_A \ln X_A + X_B \ln X_B$. This curve represents $\Delta G^{M,id}/RT$.
- ✓ Curves for $\Delta H^M/RT = \alpha X_A X_B/RT$ are drawn for $\alpha/RT = 0, +1.0, +2.0$, and $+3.0$, and the corresponding $\Delta G^M/RT$ curves are drawn as the sum of the particular $\Delta H^M/RT$ and $-(\Delta S^{M,id}/R)$ curves.



- ✓ As the magnitude of α/RT is **increased**, it is seen that the shape of the $\Delta G^M/RT$ curve continuously changes from a **shape** typified by $\alpha = 0$ to a form typified by $\alpha/RT = 3$.

Figure 10.10 the effect of the magnitude of α/RT on the reduced **enthalpies** and integral **molar Gibbs free energies** of formation of a binary regular solution.

10.5. The Gibbs Free Energy of Formation of Regular Solutions

✓ Curve I from Figure 10.1 is reproduced in Figure 10.11a. This curve has **positive curvature** at all compositions. Thus, the homogeneous solution formed from **any mixture** of A and B **is the stable state** at the temperature in question, since this state has the lowest possible Gibbs free energy.

✓ Consider, further, two separate solutions—say, **a** and **b** in Figure 10.11a.

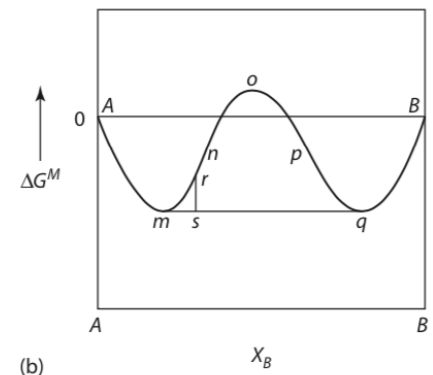
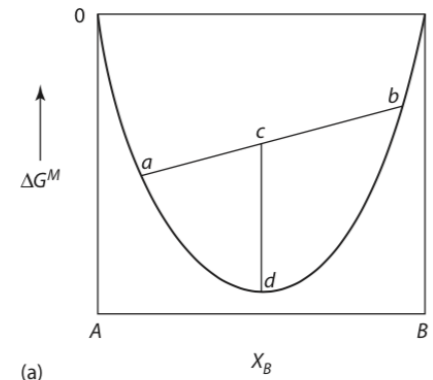


Figure 10.11

(a) the molar Gibbs free energies of mixing of binary components at a temperature which forms a complete range of solutions. (b) the molar Gibbs free energies of mixing of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ Before the mixing of these two solutions, the Gibbs free energy of the two-solution system, with respect to pure *A* and pure *B*, lies on the straight line joining **a** and **b**, with the exact position being determined, via the lever rule, by the relative proportions of the separate solutions.

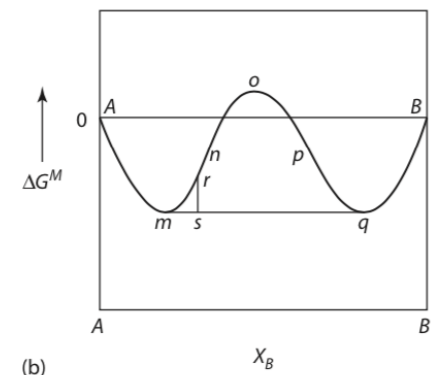
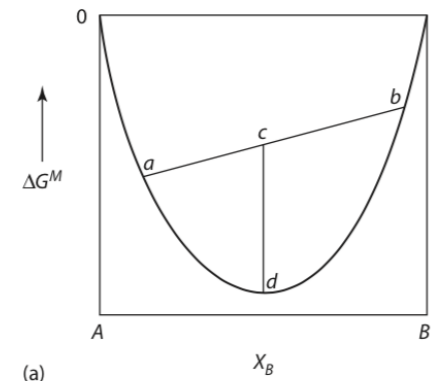


Figure 10.11

(a) the molar Gibbs free energies of mixing of binary components at a temperature which forms a complete range of solutions. (b) the molar Gibbs free energies of mixing of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ If the solutions **a** and **b** are present in equal amounts, then the Gibbs free energy of the system is given by the point **c**.
- ✓ When **mixed**, the two solutions form a new homogeneous solution, since the Gibbs free energy of the system is thereby decreased from **c** to **d**, the **minimum** Gibbs free energy of mixing it can have.

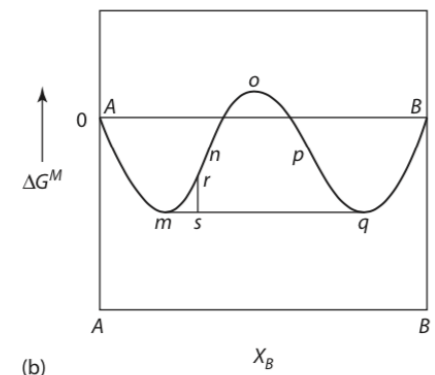
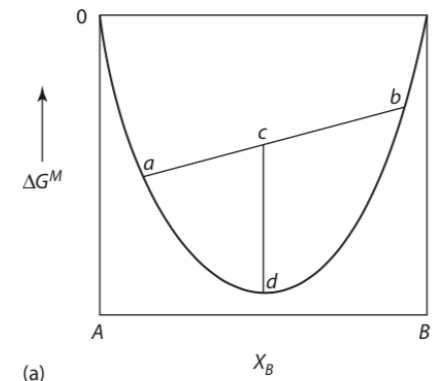


Figure 10.11

(a) the molar Gibbs free energies of mixing of binary components at a temperature which forms a complete range of solutions. (b) the molar Gibbs free energies of mixing of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ Consider now Figure 10.11b, in which the $\Delta G^M / RT$ curve for $\alpha/RT > 2$ is shown.
- ✓ This curve has **positive curvature** between **A** and **n** and between **p** and **B**, and it has **negative curvature** between **n** and **p**.
- ✓ The ΔG^M of a system of composition between **m** and **q** is **minimized** when the system occurs as two solutions, one of composition **m** and the other of composition **q**; for example,

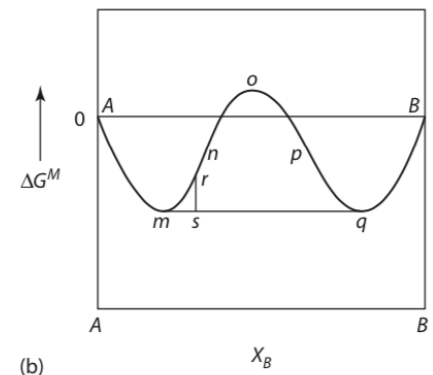
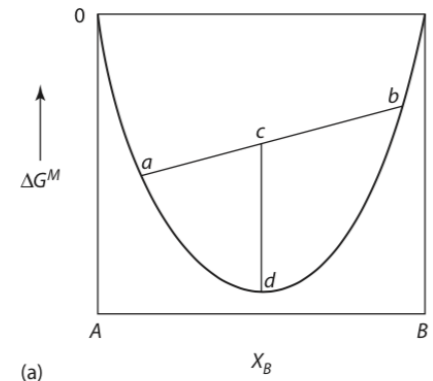


Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ if the homogeneous solution of composition r separates into the two coexisting solutions m and q , the ΔG^M of the system is **decreased** from r to s .

- ✓ The equilibrium coexistence of two separate solutions at the temperature T and pressure P requires that

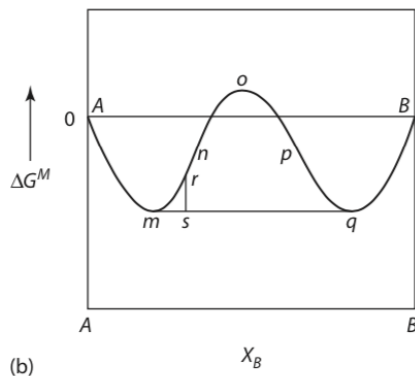
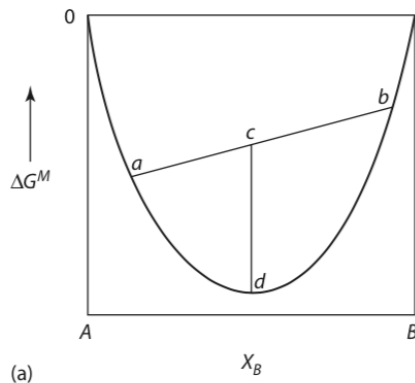
$$\overline{G}_A(\text{in solution m}) = \overline{G}_A(\text{in solution q})$$

and

$$\overline{G}_B(\text{in solution m}) = \overline{G}_B(\text{in solution q})$$

Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).



10.5. The Gibbs Free Energy of Formation of Regular Solutions

$$\overline{G}_A(\text{in solution m}) = \overline{G}_A(\text{in solution q})$$

Subtracting G_A° from both sides of Equation (i) gives

$$RT \ln \alpha_{A(\text{in solution m})} = RT \ln \alpha_{A(\text{in solution q})}$$

or

$$\alpha_{A(\text{in solution m})} = \alpha_{A(\text{in solution q})}$$

Similarly,

$$\alpha_{B(\text{in solution m})} = \alpha_{B(\text{in solution q})}$$

✓ Equation (iii) and (iv) are the criteria for the equilibrium coexistence of two solutions (or phases) at constant T and P . Since

$$\Delta \overline{G}_A^M(\text{in solution m}) = \Delta \overline{G}_A^M(\text{in solution q}) ,$$

$$\Delta \overline{G}_B^M(\text{in solution m}) = \Delta \overline{G}_B^M(\text{in solution q}) .$$

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ it is seen that the tangent to the curve at the point **m** is also the tangent to the curve at the point **q**. The positioning of this common tangent defines the positions of the points **m** and **q** on the Gibbs free energy of mixing curve.
- ✓ The *A–B* system, as represented in Figure 10.11b, is one in which, at the temperature **T**, the value of α/RT is sufficiently positive that the consequent tendency toward clustering of like atoms is great enough to cause phase separation.

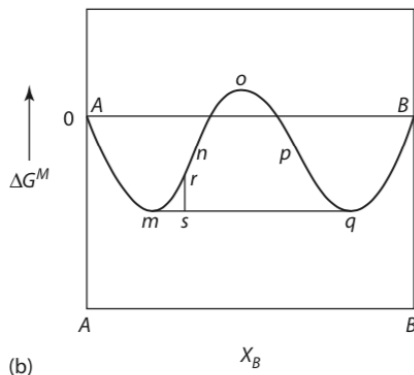


Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ A homogeneous solution (phase I) is formed when B is initially added to A and the saturation of phase I with B occurs at the composition m .
- ✓ The A – B system, as represented in Figure 10.11b, is one in which, at the temperature T , the value of α/RT is sufficiently positive that the consequent tendency toward clustering of like atoms is great enough to cause phase separation.
- ✓ A homogeneous solution (phase I) is formed when B is initially added to A and the saturation of phase I with B occurs at the composition m .

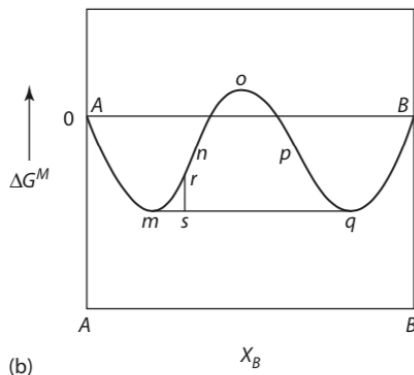


Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- ✓ Further addition of B causes the appearance of a second solution (phase II) of composition q (which is phase II saturated with A), and continued addition of B causes an increase in the ratio of phase II to phase I occurring, until the overall composition of the two-phase system reaches r , at which point phase I disappears.
- ✓ A homogeneous solution (phase II) occurs between the compositions q and B .

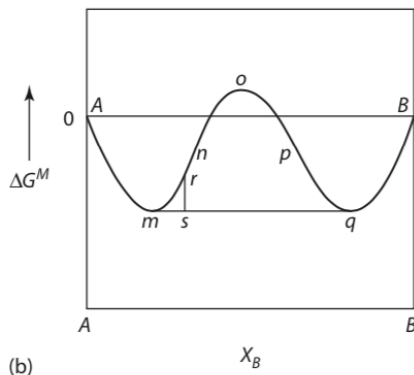


Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

10.5. The Gibbs Free Energy of Formation of Regular Solutions

- The curve **mn** represents the ΔG^M of phase I supersaturated with **B**, and
 - the curve **qp** represents the ΔG^M of phase II supersaturated with **A**.
- ✓ Since the line **AmqB** represents the equilibrium states of the system, then this line alone has physical significance, and the line is the **isobaric, isothermal** section of the system as it occurs in **G-T-P-composition** space.

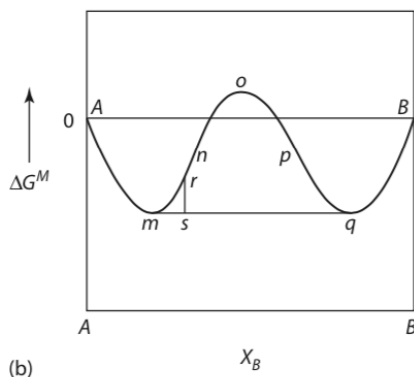


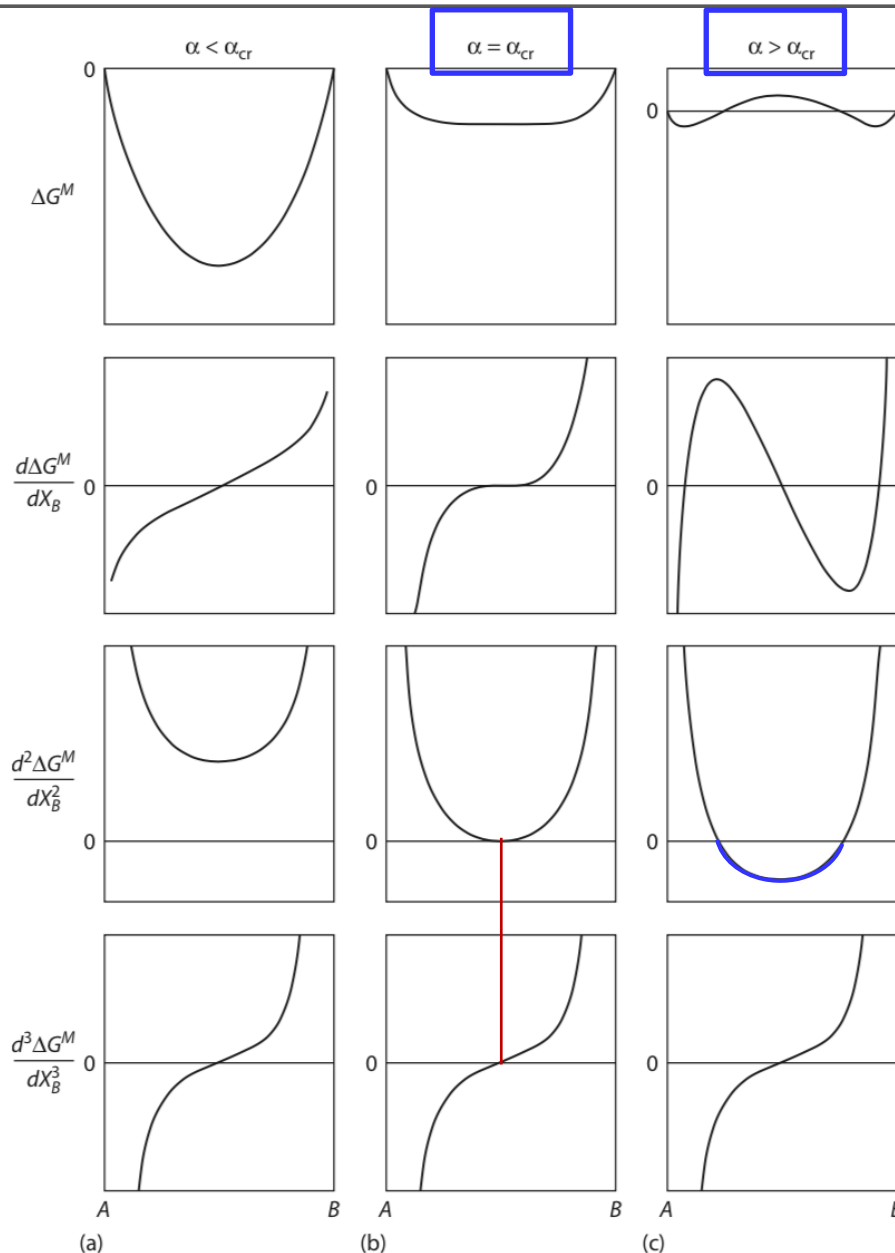
Figure 10.11

(b) the molar ΔG^M of binary components at a temperature in a system which exhibits a **miscibility gap** (two-phase region).

Contents

- ✓ 10.1 Introduction
- ✓ 10.2 Gibbs Free Energy and Thermodynamic Activity
- ✓ 10.3 Qualitative overview of Common Binary Equilibrium Phase Diagrams
- ✓ 10.4 Liquid and Solid Standard States
- ✓ 10.5 The Gibbs Free Energy of Formation of Regular Solutions
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- ✓ 10.7 Phase Diagrams, Gibbs Free Energy, and Thermodynamic Activity
- ✓ 10.8 The Phase Diagrams of Binary Systems that Exhibit Regular Solution Behavior in the Liquid and Solid States

10.6 Criteria for Phase Stability in Regular Solutions



✓ The third derivative, $\partial^3 \Delta G^M / \partial X_B^3 = 0$ at $X_A = X_B = 0.5$, and thus, the second derivative, $\partial^2 \Delta G^M / \partial X_B^2 = 0$ at $X_A = 0.5$, when $\alpha/RT = 2$, which is thus the critical value of α/RT above which phase separation occurs (Figure 10.10).

✓ Thus, the critical temperature of the miscibility gap is

$$T_\alpha = \frac{\alpha}{2R}$$

Figure 10.12 the effect of the magnitude of α on the first, **second**, and **third derivatives** of the integral Gibbs free energy of mixing with respect to composition.

10.6 Criteria for Phase Stability in Regular Solutions

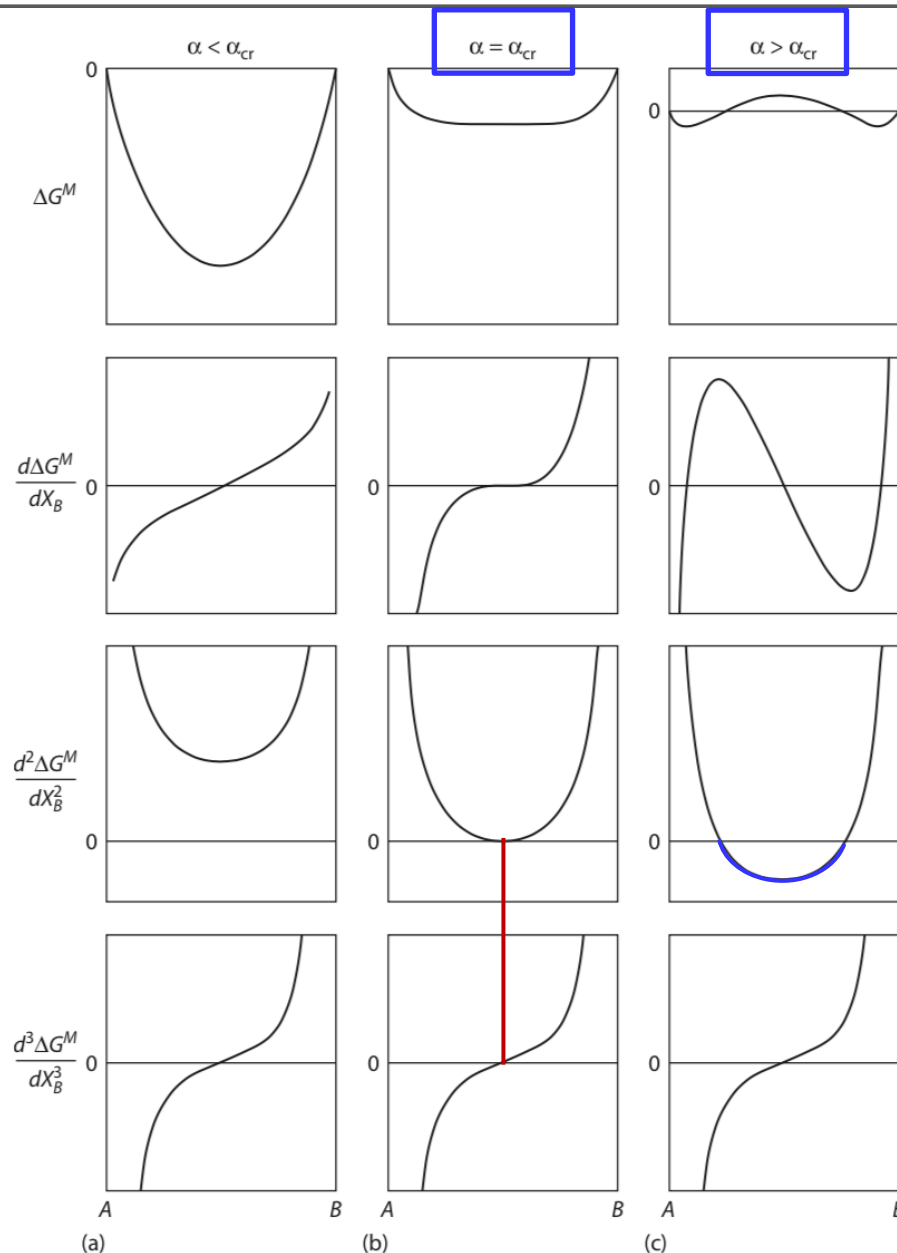
- ✓ Figure 10.10, for a given temperature, a critical value of α/RT occurs,
- ✓ below which a homogeneous solution is stable over the entire range of composition and above which phase separation occurs.
- ✓ The criteria used to determine this critical value are illustrated in Figure 10.12.
- ✓ Figure 10.12a,b, and c shows how ΔG^M , $\partial \Delta G^M / \partial X_B$, $\partial^2 \Delta G^M / \partial X_B^2$, and $\partial^3 \Delta G^M / \partial X_B^3$ vary with composition for $\alpha/RT < \alpha_{critical}/RT$, $\alpha/RT = \alpha_{critical}/RT$, and $\alpha/RT > \alpha_{critical}/RT$, respectively.
- ✓ The critical value of α/RT is seen to be that which makes $\partial^2 \Delta G^M / \partial X_B^2$ and $\partial^3 \Delta G^M / \partial X_B^3$ simultaneously equal to zero at that composition at which immiscibility becomes imminent.
- ✓ For a regular solution,

10.6 Criteria for Phase Stability in Regular Solutions

✓ For a regular solution,

$$\Delta G^M = RT(X_A \ln X_A + X_B \ln X_B) + \alpha X_A X_B, \quad \frac{\partial \Delta G^M}{\partial X_B} = RT \left[\ln \frac{X_B}{X_A} \right] + \alpha(1 - 2X_B),$$
$$\frac{\partial^2 \Delta G^M}{\partial X_B^2} = RT \left[\frac{1}{X_A} + \frac{1}{X_B} \right] - 2\alpha, \quad \frac{\partial^3 \Delta G^M}{\partial X_B^3} = RT \left[\frac{1}{X_A^2} - \frac{1}{X_B^2} \right]$$

10.6 Criteria for Phase Stability in Regular Solutions



Second, and **third derivatives**
of the integral $\Delta G^M = 0$

The critical value of α/RT :

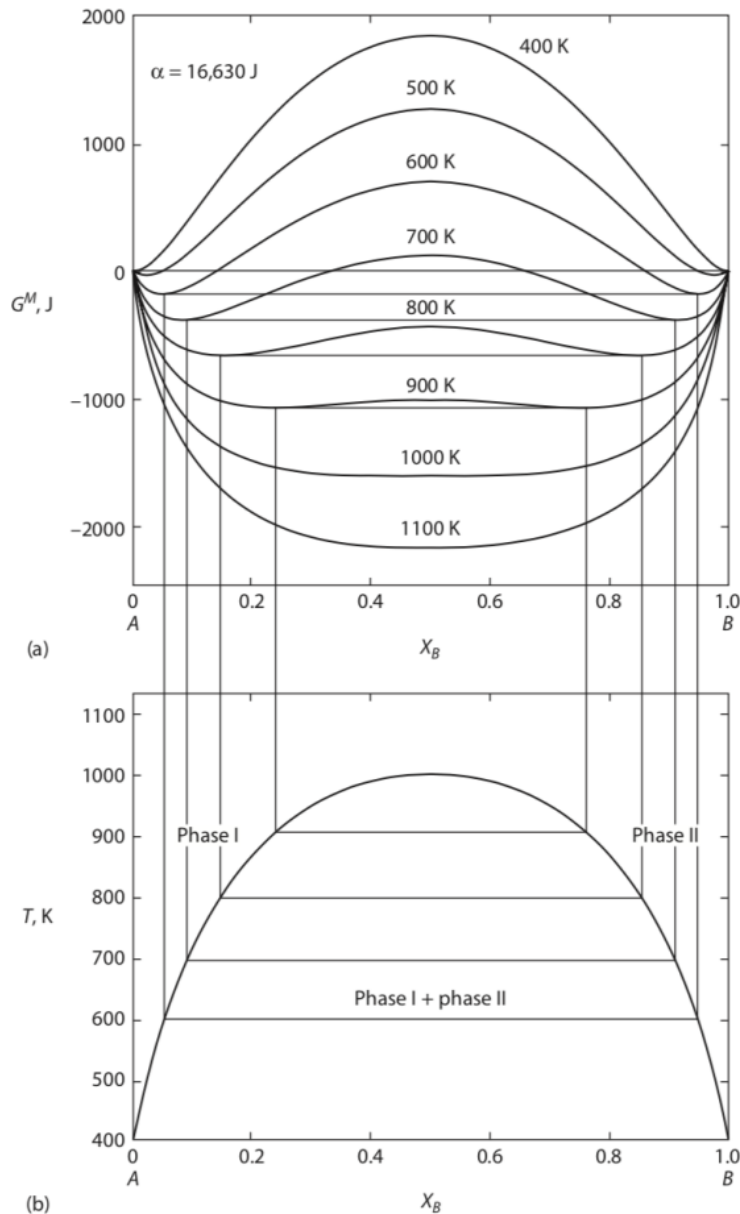
$$\partial^2 \Delta G^M / \partial X_B^2 = 0 \text{ and}$$

$$\partial^3 \Delta G^M / \partial X_B^3 = 0$$

$$\alpha/RT = 2$$

Figure 10.12 the effect of the magnitude of α on the first, **second**, and **third derivatives** of the integral Gibbs free energy of mixing with respect to composition.

10.6 Criteria for Phase Stability in Regular Solutions



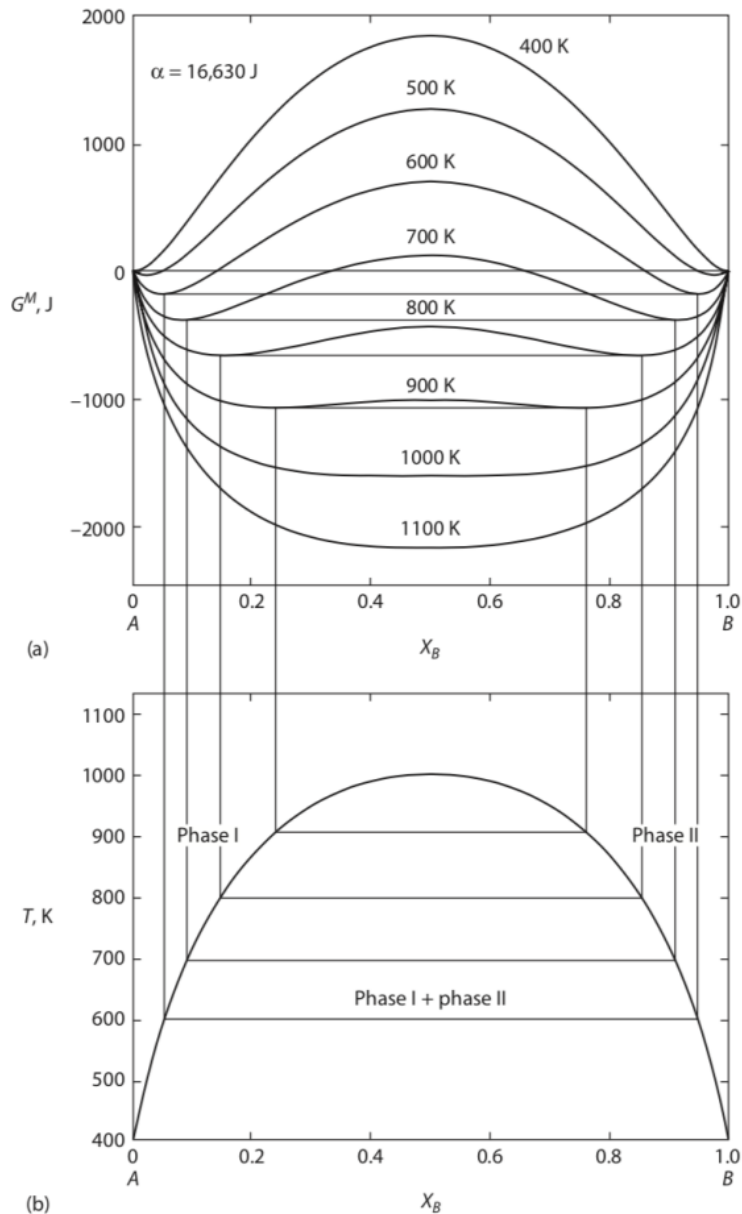
$$\Delta G^M = RT(X_A \ln X_A + X_B \ln X_B) + \alpha X_A X_B$$

- ✓ Figure 10.13a shows the variation, with temperature, of the G^M curve for a regular solution which has a positive molar heat of mixing ($\alpha=16,630$ J) and a critical temperature of $T_{cr}=16630/2R=1000$ K.

Figure 10.13

(a) the effect of temperature on the molar G^M a binary regular solution for which $\alpha = 16,630$ J. (b) the loci of the common tangent points in (a), which generate the phase diagram for the system.

10.6 Criteria for Phase Stability in Regular Solutions



- ✓ The G^M expression contains a negative logarithmic term, the magnitude of which is proportional to temperature, and a positive parabolic term which is independent of temperature.
- ✓ At high enough temperature, the logarithmic contribution predominates and the G^M has positive curvature at all compositions.

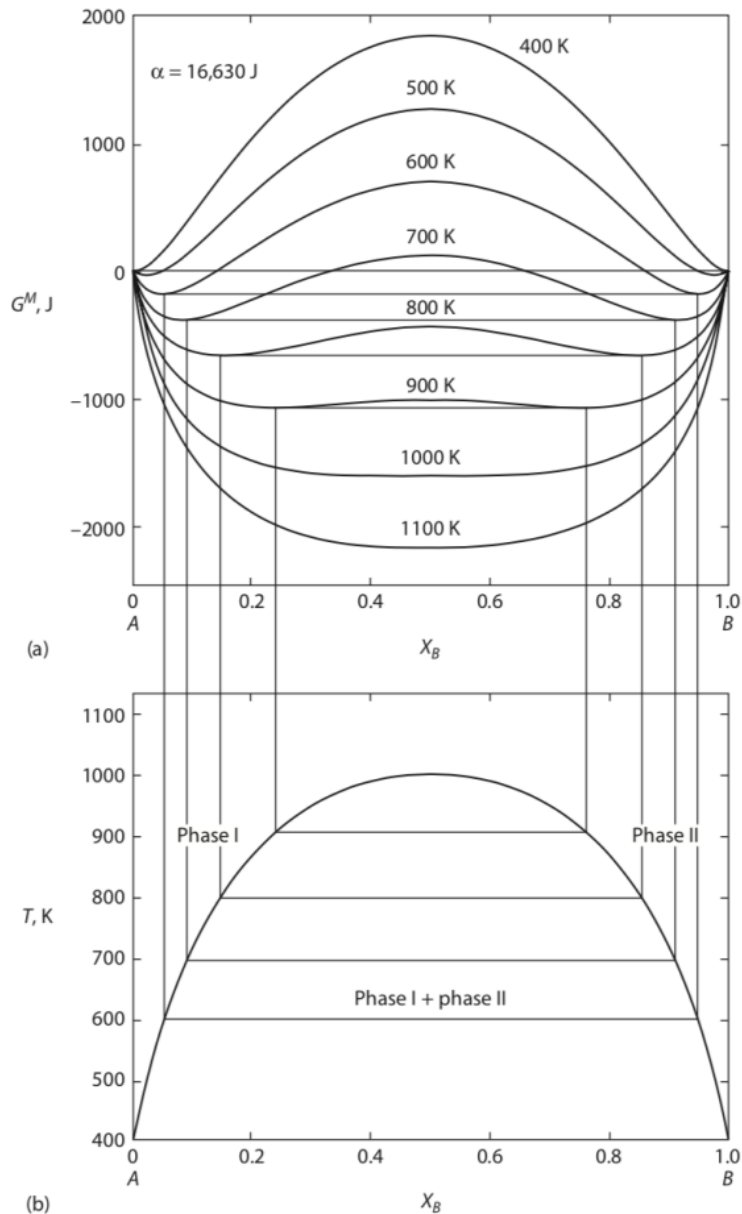
Figure 10.13

(a) the effect of temperature on the molar G^M a binary regular solution for which $\alpha = 16,630$ J. (b) the loci of the common tangent points in (a), which generate the phase diagram for the system.

10.6 Criteria for Phase Stability in Regular Solutions

- ✓ However, with decreasing temperature, the contribution of the logarithmic term decreases, and eventually, the positive parabolic term predominates and produces a range of composition centered on $X_B = 0.5$, over which the G^M curve has negative curvature.
- ✓ The logarithmic term still requires that the tangents to the curve at $X_A = 1$ and $X_B = 1$ be vertical.

10.6 Criteria for Phase Stability in Regular Solutions

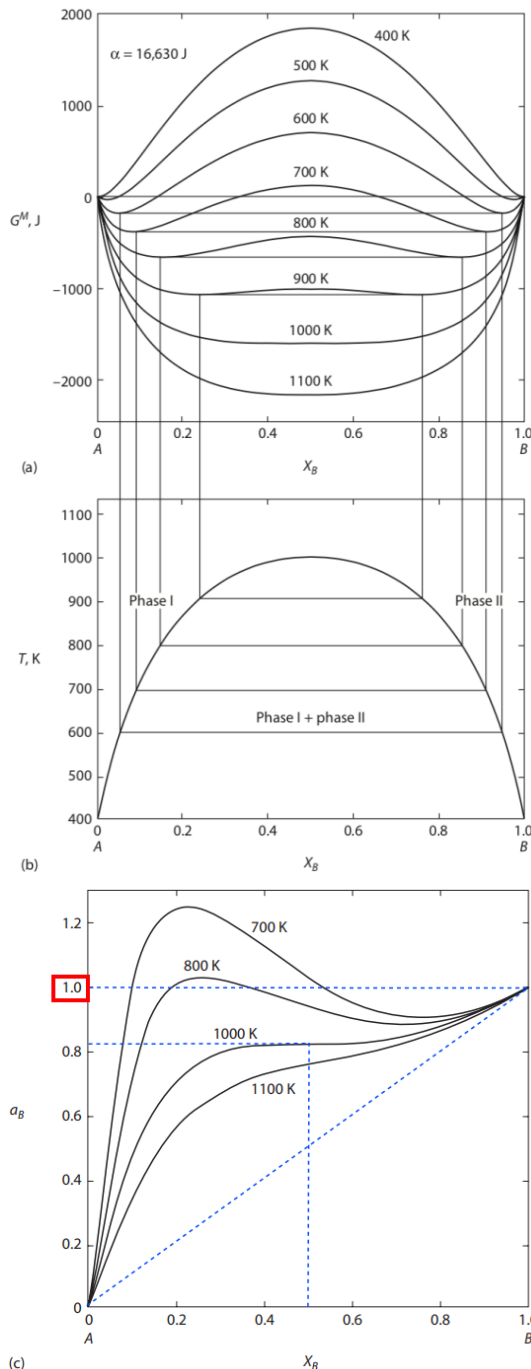


✓ Figure 10.13b shows the phase diagram for the system, in which the **miscibility curve** bounding the two phase region is simply the locus of the **common tangent** compositions in Figure 10.13a.

Figure 10.13

(a) the effect of temperature on the molar G^M a binary regular solution for which $\alpha = 16,630$ J. (b) the loci of the common tangent points in (a), which generate the phase diagram for the system.

Phase Stability in Regular Solutions



- ✓ The influence of temperature on the variations of the activity of component B with composition is shown in Figure 10.13c.
- ✓ The activities are obtained from the intercepts, with the $X_B = 1$ axis, of tangents drawn to the free energy curves as $\Delta G^M_B = RT \ln a_B$
- ✓ At T_{cr} , the activity exhibits a horizontal inflection at $X_B = 0.5$, as is seen from the following.

Figure 10.13

(a) the effect of temperature on the molar G^M a binary regular solution for which $\alpha = 16,630$ J. (b) the loci of the common tangent points in (a), which generate the phase diagram for the system. (c) the activities of component B derived from (a).

10.6 Criteria for Phase Stability in Regular Solutions

✓ From Equation (9.33b),

$$\Delta \bar{G}_B^M = \Delta G^M + X_A \left(\frac{\partial \Delta G^M}{\partial X_B} \right) = RT \ln \alpha_B$$

✓ Thus,

$$\left(\frac{\partial \Delta \bar{G}_B^M}{\partial X_B} \right) = X_A \frac{\partial^2 \Delta G^M}{\partial X_B^2} = \frac{RT}{\alpha_B} \frac{\partial \alpha_B}{\partial X_B}$$

✓ and

$$\left(\frac{\partial^2 \Delta \bar{G}_B^M}{\partial X_B^2} \right) = X_A \left(\frac{\partial^3 \Delta G^M}{\partial X_B^3} \right) - \left(\frac{\partial^2 \Delta G^M}{\partial X_B^2} \right) = \frac{RT}{\alpha_B} \frac{\partial^2 \alpha_B}{\partial X_B^2} - \frac{RT}{\alpha_B^2} \left(\frac{\partial \alpha_B}{\partial X_B} \right)^2$$

✓ At T_{cr} and $X_B = 0.5$, both the second and third derivatives of ΔG^M with respect to X_B are zero, and thus, from Equations 10.9 and 10.10, the first and second derivatives of a_B with respect to X_B are zero, which produces a **horizontal inflection** point on the activity curve at $X_B = 0.5$ and T_{cr} .

10.6 Criteria for Phase Stability in Regular Solutions

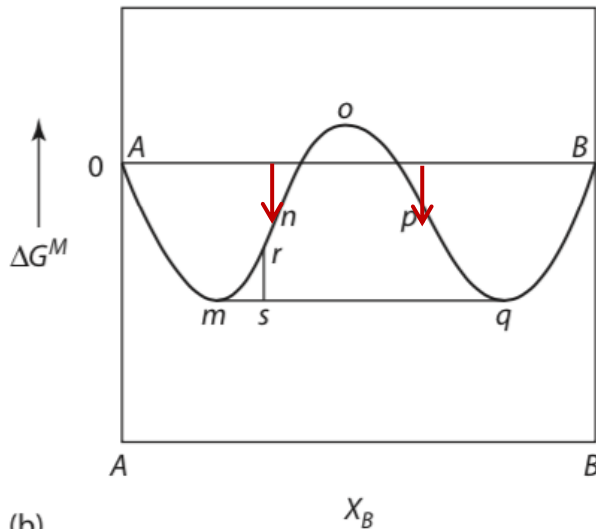
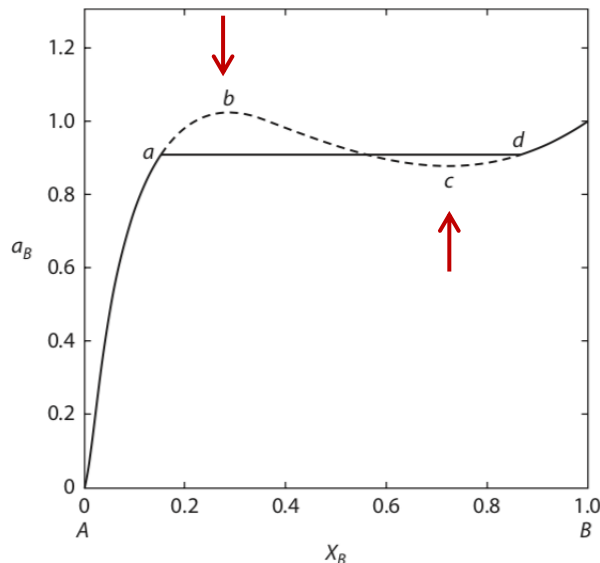


Figure 10.11 (b) the molar ΔG^M at a temperature in a system which exhibits a **miscibility gap** (two-phase region).



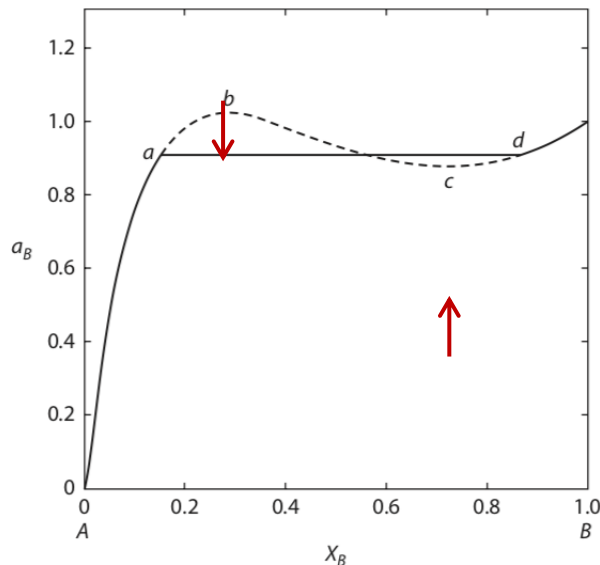
- ✓ For $T < T_{cr}$, the activity curve has a maximum and a minimum, which occur at the **spinodal compositions** (where $\partial^2 \Delta G^M / \partial X_B^2$ and hence $\partial a_B / \partial X_B$ are zero)—
- ✓ for example, the points **n** and **p** in Figure 10.11b and the points **b** and **c** on the activity curve at **800 K** shown in Figure 10.14.

- ✓ The curve **ab** represents the a_B in phase I which is supersaturated with **B**, and
- ✓ the curve **cd** represents the a_B in phase II which is supersaturated with **A**.

Figure 10.14
the activity of B at **800 K** derived from figure 10.13a.

10.6 Criteria for Phase Stability in Regular Solutions

- ✓ The value of $\partial a_B / \partial X_B$ is negative between **b** and **c**, and this **violates** an intrinsic criterion for stability, which requires that $\partial a_i / \partial X_i$ always be positive [cf. $(\partial P / \partial V)_T > 0$ over the portion *JHF* in, Figure 8.8].
- ✓ Thus, the derived activity curve between **b** and **c**, and, consequently, the ΔG^M curve between the spinodal compositions, are regions of **absolute instability**.



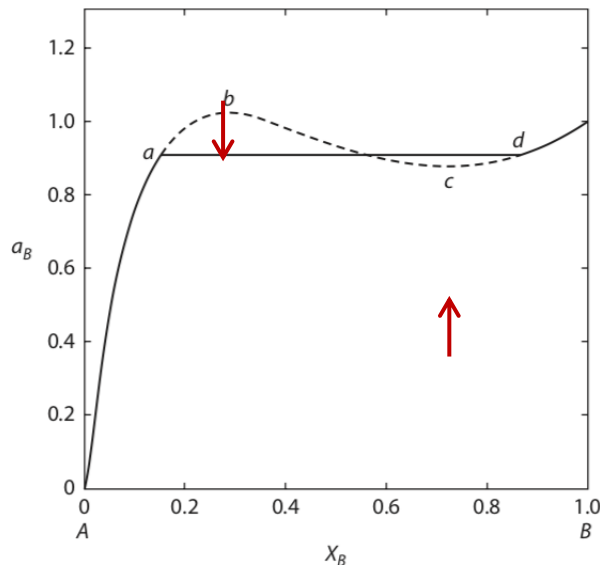
Actual constant a_B : line **ad**

X_a and X_d : double tangents to ΔG^M curve

Figure 10.14
the a_B at **800 K** derived from figure 10.13a.

10.6 Criteria for Phase Stability in Regular Solutions

- ✓ The horizontal line drawn between **a** and **d** in Figure 10.14 represents the actual constant a_B in the equilibrium two-phase region, and the compositions **a** and **d** are those of the double tangents to the ΔG^M curve.
- ✓ Thus, the regular solution under discussion has the following **different regions of stability** at 800 K ($T < T_{cr} = 1000$ K see Figure 10.14):



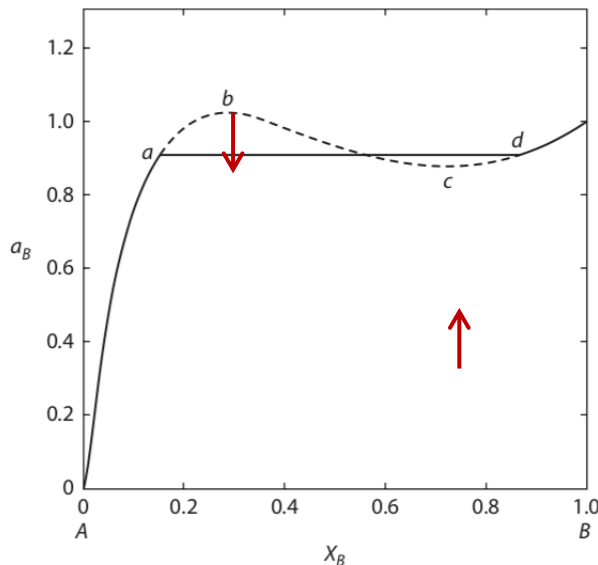
Actual constant a_B : line **ad**

X_a and **X_d**: double tangents to ΔG^M curve

Figure 10.14
the a_B at 800 K derived from figure 10.13a.

10.6 Criteria for Phase Stability in Regular Solutions

- $X_B = 0$ to $X_B = \mathbf{a}$, the single-phase solution **I** is **stable**.
- $X_B = \mathbf{a}$ to $X_B = \mathbf{d}$, the **stable** state is a **two-phase** mixture of solution I and II.
- $X_B = \mathbf{a}$ to $X_B = \mathbf{b}$, the single-phase solution of **I** is **metastable** (it will decompose into solution I of composition **a** and solution II of composition **d**).
- $X_B = \mathbf{b}$ to $X_B = \mathbf{c}$, the homogeneous solution is absolutely **unstable**.



- $X_B = \mathbf{c}$ to $X_B = \mathbf{d}$, the single-phase solution of **I** is **metastable**.
- $X_B = \mathbf{d}$ to $X_B = \mathbf{1}$, the solid solution **II** is the **i** phase.

Figure 10.14

the a_B at **800 K** derived from figure 10.13a.

Homework

4. Al_2O_3 , which melts at 2324 K, and Cr_2O_3 , which melts at 2538 K form complete ranges of solid and liquid solutions. Assuming that $\Delta S_{0m, \text{Cr}_2\text{O}_3} = \Delta S_{0m, \text{Al}_2\text{O}_3}$, and that the solid and liquid solutions in the system $\text{Al}_2\text{O}_3\text{-Cr}_2\text{O}_3$ behave ideally, calculate

The temperature at which equilibrium melting begins when an alloy of

$X_{\text{Al}_2\text{O}_3} = 0.5$ is heated,

2. The composition of the melt which first forms,

3. The temperature at which equilibrium melting is complete, and

4. The composition of last-formed solid.

Homework

5. SiO_2 , which melts at $1723\text{ }^\circ\text{C}$ and TiO_2 , which melts at $1842\text{ }^\circ\text{C}$ are immiscible in the solid state and the SiO_2 - TiO_2 binary system contains a monotectic equilibrium at $1794\text{ }^\circ\text{C}$, at which temperature virtually pure TiO_2 , is in equilibrium with two liquids containing mole fractions of SiO_2 , of 0.04 and 0.76. If, for the purpose of simple calculation, it is assumed that the compositions of the two liquids are $X_{\text{SiO}_2} = 0.24$ and $X_{\text{SiO}_2} = 0.76$ and that the liquid solutions are regular in their behavior, what is the value of Ω_l and at what temperature does the liquid immiscibility gap disappear?

Thermodynamics of Materials & Phase transformations

The End

